

Programming Fundamentals.

(Hospital Appointment Booking System.)

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1. Project Title Page

Title:

"Hospital Appointment Booking System"

2. Introduction to the Project

The Hospital Appointment System is a **C++ console-based project** designed to help patients **book appointments** with doctors easily. This system will allow users to add doctors, view doctor availability, book appointments, and view scheduled appointments. It will demonstrate the application of all basic C++ programming concepts studied in the Programming Fundamentals course.

3. Objectives of the Project

- To apply fundamental **programming concepts in C++ practically**.
- To design a real-world simple appointment system using **console-based interaction.**
- To learn how to use **variables, arrays, functions, structures, loops, and conditional statements effectively.**

4. Tools and Environment Used

- **C++**: The main language for implementing logic, functions, structures, arrays, and control flow.
- **Compiler**: Translates C++ code into executable programs.
- **Console Interface**: The project runs on the terminal/command prompt window.
- **Operating System**: Any OS that supports C++ compilers can be used.
- **Text Editor**: Optional; used if needed for easier code editing.

5. *Project Description and Features*

Main Functionality:

- **Add Doctor Information**: Admin can add doctors with name, specialization, and available timings.

- **Book Appointment**: Patients can book appointments with available doctors.

- **View Appointments**: View all the scheduled appointments along with patient and doctor details.

- **Cancel Appointment** *(Optional):* Patients can cancel an appointment if required.
- **Exit**: Close the application safely.

Key Features:

- Easy to add doctor information (ID, name, specialization, available days).
- Book appointments with doctors based on availability.
- View all booked appointments with complete details.
- Cancel an existing appointment if needed.
- Simple and user-friendly console interface.
- Organized code using structures and functions.
- Menu-driven system for smooth navigation.
- Data stored using arrays for managing multiple doctors and patients.

6. Technical Architecture and Logic

The program flow involves the following:

- **Language**: C++ (Console-based application)
- **Structures**: To store doctor and patient information
- **Static Arrays**: For managing multiple records
- **Functions**: For adding doctors, booking, viewing, and canceling appointments
- **Control Structures**: If-Else, Switch
- **Loops**: For, While, Do-While
- **Input/Output**: Using cin and cout

Logic Flow:

- Show menu
- Perform selected task
- Repeat until user exits

7. Flow of Execution

START

|

|---> Display Main Menu (Add Doctor, Book Appointment, View Appointments, Cancel Appointment, Exit)

|

|---> Loop (Until user chooses Exit):

|

|---> If Add Doctor

|---> Input doctor details

|---> Save in doctors array

|

|---> If Book Appointment

|---> Show available doctors

|---> Input patient details

|---> Save in patients array

|

|---> If View Appointments

|---> Display all appointments

|

|---> If Cancel Appointment

|---> Search and remove appointment

|

|---> Exit Program

8. Division of Work Among Members

Member	Responsibilities
Isbah	Coding doctor management (add/view doctors), Coding appointment booking and cancellation
Saher	Testing and removing errors.
Furqan	Designing the menu and user interface
All	Final documentation and project submission

9. Challenges Faced During Development

- Generating Randomness Without Standard Libraries**.**
- Converting Integers to Strings Using Only CPU Registers and Memory.
- Managing Buffer Overflows and Ensuring Clean Screen Output.
- Making the Program Modular and Readable Within the Limited Syntax of Assembly.
- Testing for Consistent Behavior Across Multiple Runs.

10. Learning Outcomes

- In-depth Understanding of Hospital Management Systems.
- Mastery Over Integration with External Systems.
- Strong Grasp of Data Security and Privacy Regulations.
- Enhanced Team Coordination and Project Management.

11. Future Scope and Enhancements

- Allow Patients to Book Appointments Online.
- Implement Waitlist Management for Fully Booked Slots**.**
- Provide Telemedicine Options for Remote Appointments.

- Enhance Doctor Availability Filters.
- Add Appointment Reminders and Notifications.

12. Conclusion

The Hospital Appointment Booking System demonstrates how technology can streamline and improve the healthcare process. By automating appointment scheduling, we not only reduced administrative workload but also provided patients with an easy, accessible way to manage their healthcare appointments. This project helped us deepen our understanding of hospital management workflows and the complexities involved in developing a secure, reliable, and user-friendly application.

13. Code Snapshot (Annotated)

Screenshots or copies of code should go here. You can label them as:

- Figure 1: Appointment Scheduling Logic.
- Figure 2: Patient Data Entry Validation.
- Figure 3: Notification System Implementation

14. References and Tools

- Hospital management system Documentation.
- HIPAA Privacy and Security Rules.
- Online API Documentation for SMS and Email Notifications.
- Class Lectures and Handouts.
- Healthcare Software Development Forums and Resources.